

Documentation (Roman Urdu)

Yeh AI Assistant ka "Network Brain" (Knowledge Graph) bnati hy, jo sirf documents read ni krtv balky alfaaz me connection dhoondti hy.

File Ka Maqsad (Purpose)

Isma SQLite ar NetworkX library k istamal sy aesa Graph (Jalla) bnya jata hy jismey misal ke tour p: [Person A] k [Company B] k sath kya taluq hy? (Entity Extractor module)

Code Blocks aur Functions ki Tafseel

1. `class EntityExtractor`

- **Kya karta hai?:** Regex expressions (Pattern matching heuristics) istml kr k user k text men sy 'Loog', 'Tareekh (Dates)', 'Companies' nikaalta ha.
- **Verbs Pattern:** Sath hi verbs dekhta hy misal: "*JavaScript is a language*" tu usko connection bna deta hy relations mn.

2. `class KnowledgeGraph (init, extract_and_store, search)`

- **Kya karta hai?:** Yeh database ma Tables bnata ha (`entities & relationships`) arr extracted graph save kar lta a.
- **`get_graph_analysis()`:** NetworkX ki madad sy Google wala **PageRank Algorithm** aur *Centrality check* run krta ha ta ky AI ko ptaa chaly sab se "Ahmii" node / object is file m kousna hj!

3. `get_related_context(query)`

- **Kya karta hai?:** RAG search k doraan ye purane paroosiyon (neighbors) nodes ko b utbaata hv takeh AI ka prompt zada azeem tar hjy!